

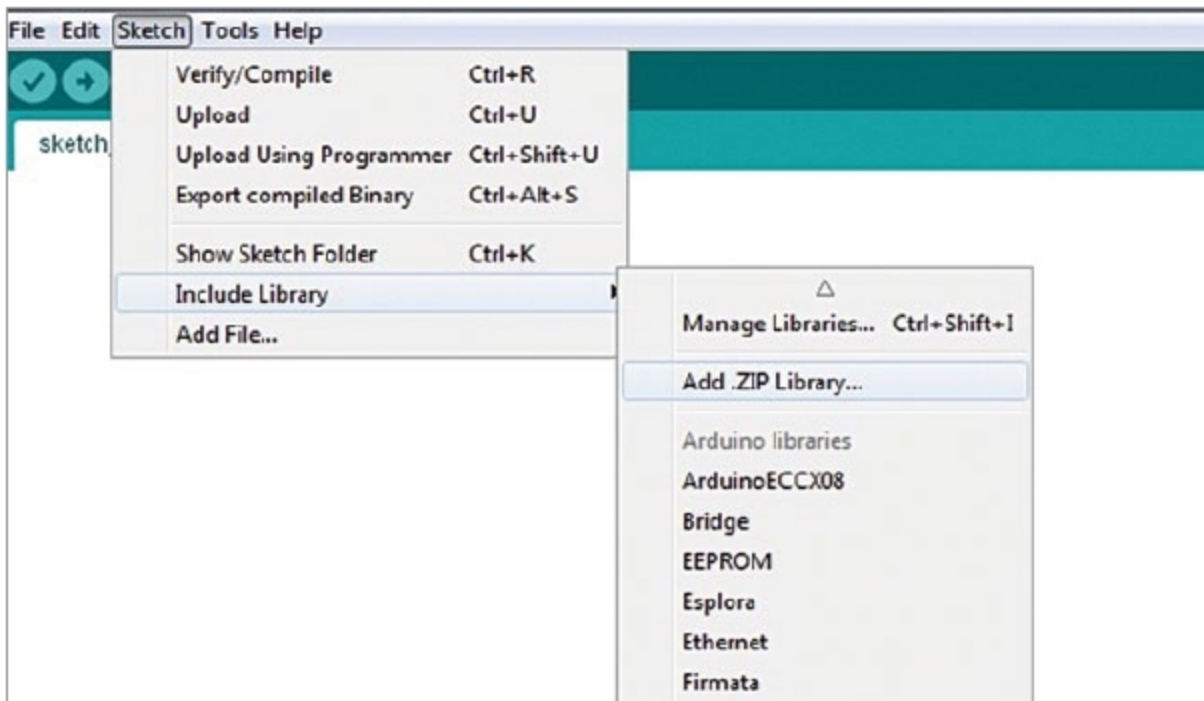


Fig. 3: Opening Library Manager

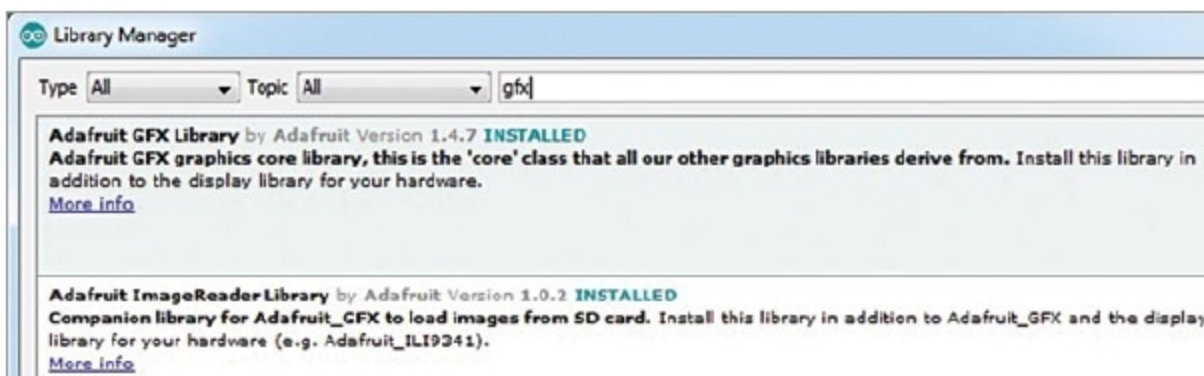


Fig. 4: Installing Adafruit GFX library

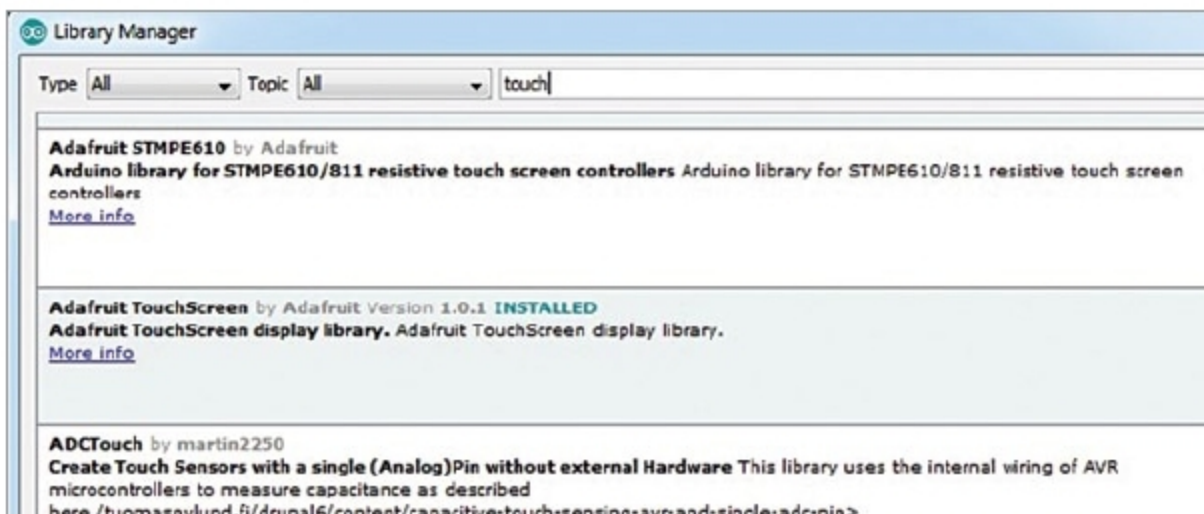


Fig. 5: Installing TouchScreen library

```
//home part
int home1=13;
int home2=12;
int home3=11;
int home4=10;
//home part
```

Next, create a function to get touch points. In this function, set the range of pressure to enable the touchscreen display to read the touch points. After that, map the values with the TFT display's width and height to identify the touch points. The code to achieve these features is given below.

```
const int XP = 6, XM = A1, YP = A2, YM = 7;
//ID=0x9341
```

```
const int TS_LEFT=901,TS_RT=123,TS_
TOP=119,TS_BOT=881;
TouchScreen ts = TouchScreen(XP, YP, XM,
YM, 300);
Adafruit_GFX_Button on_btn, off_btn, on_
btn1, off_btn1;

int pixel_x, pixel_y; //Touch_getXY()
updates global vars
bool Touch_getXY(void)
{
TSPoint p = ts.getPoint();
pinMode(YP, OUTPUT); //restore shared
pins
pinMode(XM, OUTPUT);
digitalWrite(YP, HIGH); //because TFT
```

```
control pins
digitalWrite(XM, HIGH);
bool pressed = (p.z > MINPRESSURE && p.z
< MAXPRESSURE);
if (pressed) {
pixel_x = map(p.x, TS_LEFT, TS_RT,
0, tft.width()); //kbv
makes sense to me
pixel_y = map(p.y, TS_TOP, TS_BOT,
0, tft.height());
}
return pressed;
}
```

Then, create a setup function to declare pin modes of Arduino and create buttons that you want to display on the screen of the touch display module.

In the code given below, we have created four buttons for four office rooms. We also created a few notification dots to show the working status of the bells. Colour of these notification dots will change according to status. For example, it will turn red when the office is closed, and green when it is open. The code to achieve these features is given below.

```
void setup(void)
{
Serial.begin(9600);
pinMode(home1,OUTPUT);
pinMode(home2,OUTPUT);
pinMode(home3,OUTPUT);
pinMode(home4,OUTPUT);
uint16_t ID = tft.readID();
if (ID == 0xD3D3) ID = 0x9486;
// write-only shield
tft.begin(ID);
tft.setRotation(0); //PORTRAIT
tft.fillRect(BLACK); //left.up.size.
size
tft.setFont(&FreeMonoBold9pt7b);
on_btn.initButton(&tft, 105, 90, 190,
40, GREE, BLACK, GREE, "Office 1", 1);
off_btn.initButton(&tft, 105, 140, 190,
40, GREE, BLACK, GREE, "Office 2", 1);
on_btn1.initButton(&tft, 105, 190, 190,
40, GREE, BLACK, GREE, "Office 3", 1);
off_btn1.initButton(&tft,105, 240, 190,
40, GREE, BLACK, GREE, "Office 4", 1);
on_btn.drawButton(false);
off_btn.drawButton(false);
on_btn1.drawButton(false);
off_btn1.drawButton(false);
}
```

Now, create a function to check and return the status of the buttons, that is, whether touch buttons are pressed or not. The code to achieve these features is given below.

```
Adafruit_GFX_Button *buttons[] = {&on_btn,
```